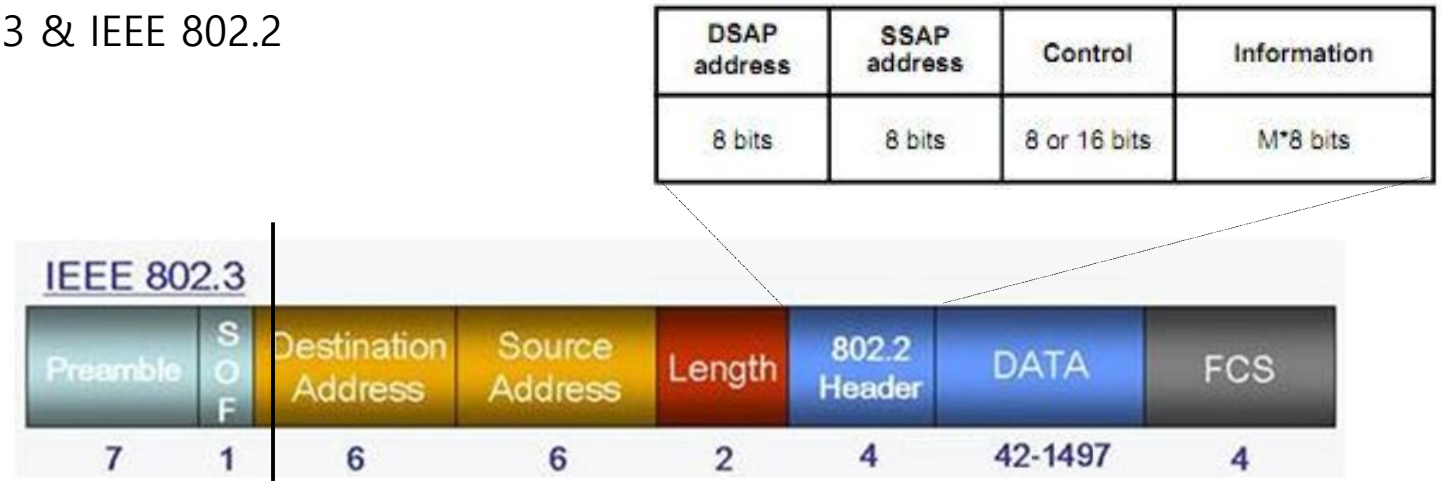
ARP cache table update

Ethernet Frame 구조

❶ Ethernet Frame (DIX 2.0)

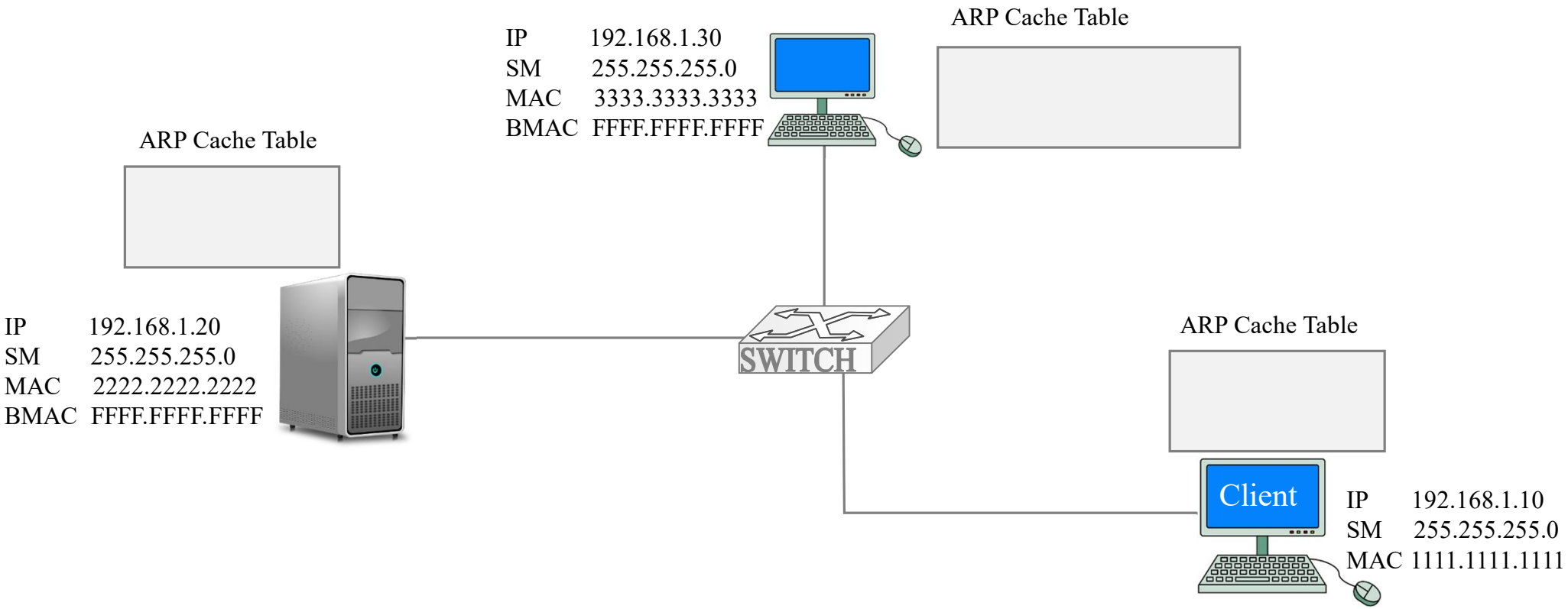
❷ IEEE 802.3 & IEEE 802.2



ARP 캐시 테이블(Cache Table)

- IP주소와 MAC 주소의 대응 관계를 저장한 테이블
- ARP 캐쉬 테이블 확인 명령어 : arp -a

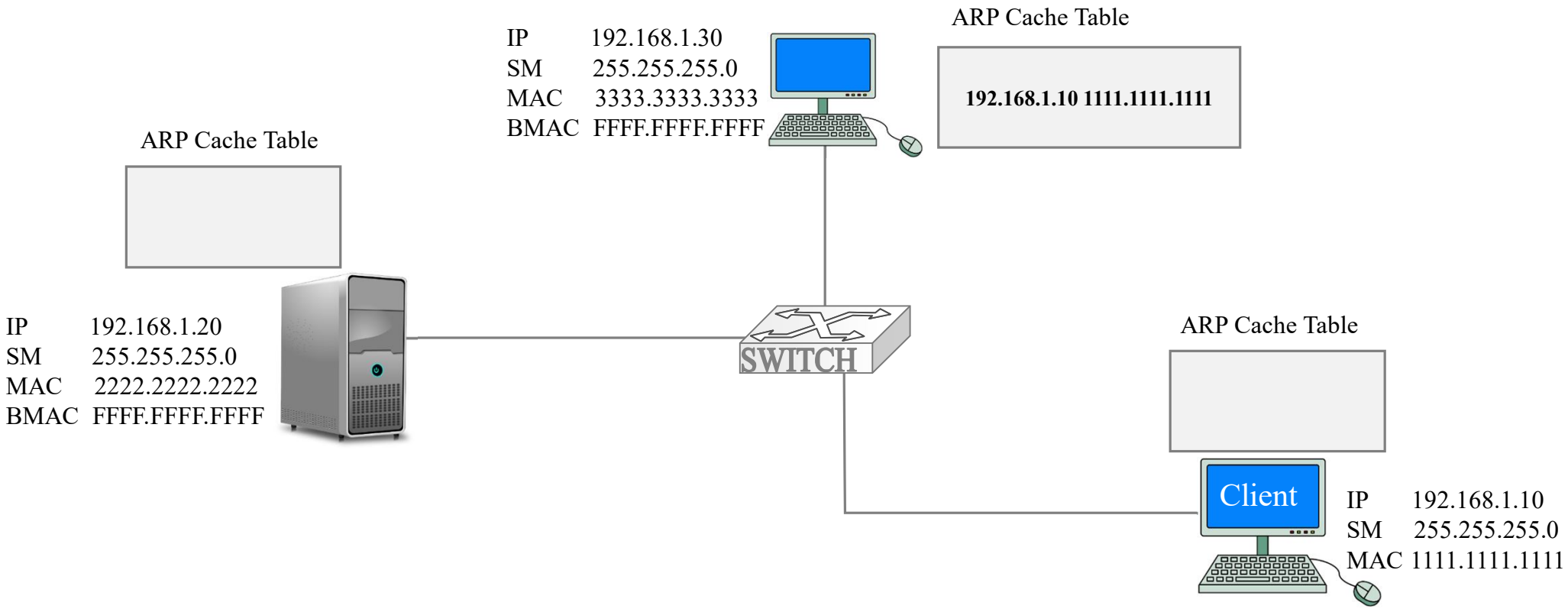
```
C:₩>arp -a

인터페이스: 192.168.35.131 --- 0x4
  인터넷 주소           물리적 주소           유형
  192.168.35.1          00-23-aa-83-11-69     동적
  192.168.35.115        38-8c-50-9a-9b-b1     동적
  192.168.35.211        04-b4-29-bf-07-09     동적
  192.168.35.255        ff-ff-ff-ff-ff-ff     정적
  224.0.0.22            01-00-5e-00-00-16     정적
  224.0.0.251           01-00-5e-00-00-fb     정적
  224.0.0.252           01-00-5e-00-00-fc     정적
  239.255.255.250       01-00-5e-7f-ff-fa     정적
  255.255.255.255       ff-ff-ff-ff-ff-ff     정적
```



ARP Request

수MAC	수MAC	Protocol	ARP Header	
FFFF.FFFF.FFFF	1111.1111.1111	0x0806	1111.1111.1111 /192.168.1.10 **0000.0000.0000/192.168.1.20**	송신지 수신지

수MAC	송MAC	송IP	수IP	송Port	수Port	전송데이타
????	1111. 1111.1111	192.168.1.10	**192.168.1.20**	50030	80	Get /www.test.com



ARP Request

수MAC	수MAC	Protocol	ARP Header
FFFF.FFFF.FFFF	1111.1111.1111	0x0806	1111.1111.1111 /192.168.1.10 송신지 **0000.0000.0000/192.168.1.20 수신지**

수MAC	송MAC	송IP	수IP	송Port	수Port	전송데이타
????	1111. 1111.1111	192.168.1.10	**192.168.1.20**	50030	80	Get /www.test.com

	ARP Header	Protocol	송MAC	수MAC
송신지 수신지	2222.2222.2222 /192.168.1.20 1111.1111.1111 /192.168.1.10	0x0806	2222.2222.2222	1111.1111.1111

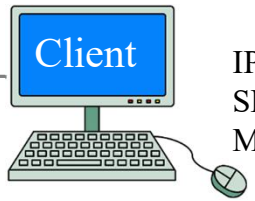
ARP Cache Table

192.168.1.10 1111.1111.1111

IP 192.168.1.20
SM 255.255.255.0
MAC 2222.2222.2222
BMAC FFFF.FFFF.FFFF

ARP Cache Table

192.168.1.20 2222.2222.2222



IP 192.168.1.10
SM 255.255.255.0
MAC1111.1111.1111

수MAC	송MAC	송IP	수IP	송Port	수Port	전송데이타
222.2222.2222	1111. 1111.1111	192.168.1.10	192.168.1.20	50030	80	Get /www.test.com